

BROKEN HILL, AustraliaWMO: **946910**Lat: **32.0011S**Lon: **141.4694E**Elev: **282**StdP: **97.99**Time Zone: **9.50 (AUA)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.1	3.3	-8.6	1.9	22.4	-6.8	2.2	22.6	11.7	14.1	10.6	13.6	1.9	210	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	40.0	18.8	38.1	18.4	36.2	18.0	22.4	29.3	21.3	29.6	20.3	30.2	5.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.9	24.5	19.1	14.4	23.8	17.3	12.8	23.9	67.4	28.7	63.1	29.7	59.4	29.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.2	9.2	-0.2	42.1	0.9	1.9	-0.9	43.4	-1.4	44.5	-1.9	45.5	-2.6	46.9		
			DB													
			WB	-1.1	23.8	1.0	1.3	-1.8	24.8	-2.4	25.5	-3.0	26.3	-3.7	27.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	26.7	25.7	22.1	17.9	13.6	10.7	10.3	11.7	15.3	18.7	21.8	24.5
	DBStd	6.90	4.65	4.47	4.35	3.78	3.13	2.68	2.51	3.04	3.89	4.31	4.75	4.62
	HDD10.0	67	0	0	0	0	3	21	27	14	1	1	0	0
	HDD18.3	1082	1	2	13	53	151	228	251	206	109	49	16	4
	CDD10.0	3064	517	440	376	237	115	43	34	68	160	269	355	450
	CDD18.3	1037	259	208	130	41	4	1	0	2	18	59	121	195
	CDH23.3	13718	3642	2736	1536	424	31	1	1	20	237	789	1601	2699
	CDH26.7	7197	2154	1514	711	128	1	0	0	2	67	323	810	1487
Wind	WSAvg	4.8	5.5	5.3	4.8	4.2	4.0	4.0	4.3	4.5	4.9	5.2	5.3	5.4
Precipitation	PrecAvg	266	32	14	22	26	25	14	17	20	25	29	17	25
	PrecMax	714	166	82	234	216	84	61	51	74	136	109	66	106
	PrecMin	70	0	0	0	0	0	1	0	0	0	0	0	0
	PrecStd	125	38	19	43	41	26	14	13	17	32	30	16	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	41.6	37.7	32.7	26.1	22.2	21.7	25.7	31.2	35.1	39.8	41.0
		MCWB	19.4	20.4	18.3	16.4	14.1	13.1	11.8	12.7	14.7	16.1	18.1	18.4
	2%	DB	40.5	38.8	35.3	29.9	24.0	19.8	19.2	22.7	28.7	32.8	36.5	38.6
		MCWB	19.0	19.0	17.9	15.7	13.6	12.1	10.8	11.7	13.7	15.4	17.1	17.9
	5%	DB	38.3	36.4	33.1	27.9	21.9	17.7	17.3	20.5	26.1	30.4	33.9	36.4
		MCWB	18.9	18.7	17.4	14.8	12.8	11.1	10.1	10.7	13.0	14.7	16.3	17.3
	10%	DB	36.1	34.4	30.7	25.7	19.9	16.0	15.8	18.5	23.5	27.8	31.4	34.1
		MCWB	18.4	18.3	16.9	14.5	12.3	10.6	9.5	10.1	12.0	13.8	15.7	17.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	24.1	21.9	18.9	16.5	15.1	14.6	14.6	16.8	18.3	20.8	22.5
		MCDB	29.4	29.2	28.3	25.3	20.8	19.1	17.9	21.2	24.4	27.0	28.4	28.6
	2%	WB	22.0	22.8	20.6	17.6	15.2	13.5	12.1	12.9	15.1	17.1	19.5	20.5
		MCDB	29.7	29.0	27.8	23.9	20.3	17.6	15.8	20.1	24.6	26.4	27.9	29.5
	5%	WB	21.0	21.6	19.4	16.5	14.1	12.2	11.0	11.6	14.0	15.9	18.4	19.3
		MCDB	30.9	29.7	28.3	24.2	19.7	16.4	15.6	19.0	23.6	27.2	29.1	30.4
	10%	WB	20.0	20.3	18.2	15.3	13.1	11.1	10.1	10.7	12.9	14.9	17.3	18.3
		MCDB	31.9	30.2	28.0	23.9	18.8	15.0	14.7	17.1	21.7	25.5	28.4	31.2
Mean Daily Temperature Range	5% DB	MDBR	13.5	12.7	12.6	12.2	10.6	9.5	9.8	11.5	12.6	13.2	13.2	13.6
		MCDBR	15.2	14.3	14.4	14.3	12.6	11.2	11.8	13.9	15.4	16.0	15.5	15.2
		MCWBR	5.7	5.3	5.8	6.1	5.9	5.9	6.2	6.5	6.4	6.6	6.0	5.6
	5% WB	MCDBR	12.3	10.6	11.4	11.3	10.0	9.3	9.8	12.0	13.0	12.9	12.8	12.4
	MCWBR	5.4	4.3	5.2	5.3	5.3	5.4	5.7	6.2	6.1	6.0	5.2	5.5	
Clear-Sky Solar Irradiance	taub	0.329	0.328	0.311	0.300	0.287	0.279	0.281	0.291	0.328	0.325	0.336	0.326	
	taud	2.507	2.535	2.569	2.571	2.582	2.592	2.574	2.527	2.411	2.441	2.448	2.488	
	Ebn at Noon	1010	991	976	935	898	882	896	930	940	983	996	1016	
	Edh at Noon	114	107	97	86	76	70	75	88	111	116	120	117	
All-Sky Solar Radiation	RadAvg	8.11	7.33	6.22	4.76	3.49	2.90	3.20	4.20	5.62	6.87	7.54	8.14	
	RadStd	0.45	0.39	0.40	0.23	0.19	0.16	0.15	0.22	0.32	0.34	0.41	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page